Kaung Htet

Data Engineer | AI/ML Engineer | Software Engineer
Singapore | +65 9342 0524 | lewisnick.ln8899@gmail.com | linkedin.com/in/kaung-htet-benjamin | github.com/KaungOrYours | kaung-htet-portfolio.vercel.app

Summary

Data and AI engineer, 4+ years across data pipelines, ML systems and industrial software. Currently Lead Software Developer and AI Engineer: C#/.NET control software on Siemens S7 PLCs, a Python/SQL/InfluxDB time-series layer, and an on-premise AI document-intelligence platform. First-author IEEE Xplore publication; First Class BSc; MSc Data Science (in progress).

Technical Skills

Languages & Backend: Python, SQL, C#, JavaScript, Bash, FastAPI, Flask, REST API design, pytest, .NET/WinForms, PySide6
Data & ML: ETL/ELT, data pipelines, data modelling, Databricks, AWS Glue, pandas, NumPy, time-series, InfluxDB, SQLite, scikit-learn, PyTorch, XGBoost, ensembles, SHAP, TF-IDF, recommender systems, OCR
AI & LLM: LangChain, Ollama/Llama 3.2, Claude API, Model Context Protocol (MCP), multi-agent orchestration, RAG, prompt engineering, LLM output validation, agent-based modelling (Mesa), Monte Carlo
Cloud, BI & Industrial: Docker, AWS (EC2, Glue), Vercel, RunPod, Linux, Git, CI/CD, n8n, Power BI, Tableau, Grafana, Siemens S7-1500 (S7netplus), Modbus RTU, DXF/DWG (ezdxf, ACadSharp)

Professional Experience

Lead Software Developer and AI Engineer | *Tiancheng Electrical Marine Equipment, Singapore* Aug 2026 – Present

- Own the full stack for offshore vessel positioning and automatic heave compensation: C# (.NET) WinForms control applications on Siemens S7-1500 PLCs via S7netplus plus a dynamic positioning system over Modbus RTU; deployed live on a rock-dumping barge (Sep 2026).
- Built and run the time-series data layer – SQLite, InfluxDB v1/v2 with retention policies, Telegraf replication to AWS EC2, and Grafana dashboards for real-time sensor and winch telemetry.
- Technical lead on an 8-month AI document-intelligence programme (3 pilot product lines, 7 deliverables): Python DXF/DWG-to-JSON pipeline (ezdxf, ACadSharp, pytest, PySide6) cutting round-trip entity loss to 0.72% and resolving 4 AutoCAD compatibility defects; automated labelling produced 1,200+ validated component records, replacing manual annotation.
- Architected the on-premise AI environment (Python MCP server, 10 specialist agents, file-based memory, signed-commit Git) – a zero-cloud design meeting a 7-control client security baseline. Manage releases/rollback to industrial PCs; lead junior developers.

Freelance Software and Data Engineer | *Independent, Singapore / Remote* Jan 2024 – Jul 2026

- Delivered end-to-end data products for SME and consulting clients: FastAPI backends, Python processing layers and web frontends, containerised with Docker Compose and shipped to cloud hosting.
- Automated recurring client workflows with n8n on Docker (scheduled ingestion, webhook pipelines) and processed large datasets on RunPod GPU cloud through reproducible, version-controlled pipelines feeding interactive dashboards.

Data Analyst | *Khine Myal Min Co. Ltd., Mandalay, Myanmar* Apr 2021 – Apr 2023

- Built Python and SQL pipelines across sales, supply-chain and operations data, automating extraction, transformation and reporting and cutting data-preparation time from hours to minutes.
- Built and maintained live Power BI dashboards on SQL sources, used daily by management and operations to drive stock, pricing and distribution decisions.

Selected Projects and Publication

Ka-Nova Hybrid Generative Agent-Based Model | Python, Mesa, LangChain, Llama 3.2, RunPod 2026
MSc dissertation engine: 11,000 rule-based Mesa citizen agents plus 7 LLM elite decision agents per scenario over a 17-vector external world. 300 Monte Carlo runs, 1.5M simulated agent-years on RTX 4090; results separated at $p < 0.001$, Cohen's $d \geq 1.21$. Built CVES, a 4-layer runtime pipeline validating every LLM decision before it enters the simulation.

Conflict Risk Forecasting – published, IEEE Xplore | Python, scikit-learn, SHAP, SQL, Tableau 2025
Reproducible ETL and ML pipeline harmonising UCDP, World Bank and Freedom House into a 12,700-record country-year dataset (195 countries, 1960–2023). Soft-voting ensemble: 0.95 AUC, 0.82 accuracy, 0.81 F1 on the 2023 hold-out, with SHAP interpretability and a published Tableau dashboard. **First author**, IEEE RASSE 2025 (DOI: 10.1109/RASSE64831.2025.11315309).

Recipe and Wine Pairing Recommender | TF-IDF, scikit-learn, pandas – content-based recommender over 831k merged recipes (fuzzy de-duplicated to 398k) and 1.13M user interactions, with a 35k-row wine-food pairing join. **PDF to EPUB Platform** | FastAPI, Tesseract OCR, PyMuPDF, Docker – auto-detects scanned versus digital PDFs, multilingual OCR (English, Burmese, Chinese), heuristic table-of-contents detection, async submit/poll/download job API.

Education and Awards

MSc Data Science – University of Hertfordshire (PSB Academy), Singapore; top modules 91, 84 Jan – Dec 2026
BSc (Hons) Computing Science, First Class Honours – Coventry University (PSB Academy) 2024 – 2025
HND Computer Programming, Grade A – iNet College, Mandalay, Myanmar 2018 – 2023
Awards: PSB "Student Shine on Global Stage" 2025; Pearson Overall Merit Award; Google Data Analytics Certificate | **Languages:** English, Burmese